Cashous W. Bortner

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Employment

Associate Professor of Mathematics [Tenured] **August 2026 - Present**
California State University, Stanislaus

Assistant Professor of Mathematics **August 2022 - August 2026**
California State University, Stanislaus

Research Interests

Algebraic, combinatoric, and graph theoretic approaches to problems in biological and physical systems; parameter identifiability; algebraic statistics.

Education

Ph.D. in Mathematics, North Carolina State University **May 2022**
Thesis: "Identifiability Analysis of Two Families of ODE Models" advised by Seth Sullivant

M.S. in Mathematics, North Carolina State University **May 2019**

B.S. in Mathematics, University of Nebraska-Lincoln **May 2017**

Publications

1. **Bortner**, **C.** *et al.* Maximum Likelihood Degree of the β-Stochastic Blockmodel. *Algebraic Statistics* **16,** 77–94. https://msp.org/astat/2025/16-1/p04.xhtml (2025).
2. **Bortner**, **C.** & Meshkat, N. Graph-Based Sufficient Conditions for the Indistinguishability of Linear Compartmental Models. *SIAM Journal on Applied Dynamical Systems* **23,** 2179–2207. https://doi.org/10.1137/23M1614663 (2024).
3. **Bortner**, **C.**, Gross, E., Meshkat, N., Shiu, A. & Sullivant, S. Identifiability of Linear Compartmental Tree Models and a General Formula for the Input-Output Equations. *Advances in Applied Mathematics* **146.** https://doi.org/10.1016/j.aam.2023.102490 (May 2023).
4. **Bortner**, **C.** & Sullivant, S. Structural Identifiability of Series-Parallel LCR Systems. *Journal of Symbolic Computation.* **112,** 79–104. https://doi.org/10.1016/j.jsc.2022.01.002 (Sept. 2022).
5. **Bortner**, **C.** & Meshkat, N. Identifiable Paths and Cycles in Linear Compartmental Models. *Bulletin of Mathematical Biology.* **84,** 53. https://doi.org/10.1007/s11538-022-01007-5 (Mar. 2022).
6. Bernstein, P., **Bortner**, **C.**, Coskey, S., Li, S. & Simpson, C. The Set Splittability Problem. *The Australasian Journal of Combinatorics* **75,** 190–209. https://ajc.maths.uq.edu.au/pdf/75/ajc_v75_p190.pdf (Oct. 2019).

In Preparation

1. **Bortner, C.**, Chamorro, E., & Ruvalcaba Gómez, I. Indistinguishability of Directed Cycle Models. (Expected Submission Fall 2026.)

2. **Bortner, C.**, Campos, P., De Silva, J. & Venable, J. Maximum Stars in the Corona of Ladder Graphs. (Expected Submission Fall 2026).

3. **Bortner, C.**, Gilliana, J., Patel, D., & Tamras, Z. Graph Theoretic Proofs of Linear Compartmental Model Indistinguishability. (Submitted.) Preprint available at https://arxiv.org/abs/2412.01135

Undergraduate Research Advisor

Indistinguishability of Linear Compartmental Tree Models Fall 2026-Present
Tatiana Cherukuri (Math, funded by NSF LEAPS grant), Nicole Little (Math, funded by NSF LEAPS grant), and Rene Rodriguez (Math, funded by NSF LEAPS grant);

Identifiability of Linear Compartmental Path Models Fall 2025-Present
Ariana Puccini (Math/CS, funded by NSF LEAPS grant), Emily Olivares (Math, funded by NSF LEAPS grant), and Emanuel Vega-Maldonado (CS, funded by NSF LEAPS grant); Poster Presentations: 2026 Stan State College of Science Research Celebration

Introduction to Robotics in Agriculture Fall 2025-Present
Jacky Sierra Lopez (Math, funded through STEM-CRU), Leonardo Morales (Math), and Cooper Boone (Math); Poster Presentations: 2026 Stan State College of Science Research Celebration

Graph Shelling as a Measure of Centrality Fall 2025-Spring 2026
Joseph Reynosa (Math) and Emmanuel Mendoza Rosales (CS)
Poster Presentations: 2026 Stan State College of Science Research Celebration

Indistinguishability of Linear Compartmental Models Spring 2025-Spring 2026
Eduardo Chamorro (Math, funded through LSAMP, McNair, and NSF LEAPS grant) and Ingrid Ruvalcaba Gomez (Math/Sociology, funded through NSF LEAPS grant)
Presentations: 2025 Stan State Capstone Conference [poster], 2025 UCLA National McNair Conference (UCLA) [poster], 2025 CSU Summer Symposium & Graduate Program Fair at UCLA (UCLA) [poster], 2026 Joint Mathematics Meetings (Washington DC) [poster], CSUBIOTECH Annual Symposium (Sacramento, CA) [poster], Stan State Undergraduate Research Competition [presentation], 2026 Stan State College of Science Research Celebration [poster], Central Valley Research Symposium (Fresno, CA) [presentation], Math for All (Corvallis, OR) [presentation], 2026 Stan State Capstone Conference [presentation].

Data Science in Chemistry Fall 2024-Spring 2025
Co-Advised with Tyler Knapp (Moses Lake Industries)
Francisco Duran (CS, funded through LSAMP) and Jacky Sierra Lopez (Math, funded through STEM-CRU)

Global Identifiability Testing of Linear Compartmental Tree Models Fall 2024-Spring 2025
Tana Randolph (Math, funded through LSAMP) and Omar Rios Cruz (CS, funded through STEM-CRU)

Graph Theoretic Proofs of Linear Compartmental Model Indistinguishability Spring 2024 - Fall 2024
John Gilliana (Math/Economics), Dev Patel (Math), and Zaia Tamras (Math)

Indistinguishability of Linear Compartmental Cycle Models Fall 2023
Dev Patel (Math) and Jennifer Ayson (Math, funded through LSAMP)

Erdös-Ko-Rado Theorem on Graphs Spring 2023 - Fall 2023
Co-Advised with Jessica De Silva (CSU Stanislaus)
Paola Campos (Math) and Jeffrey Venable (Math)
Poster Presentations: 2024 Joint Mathematics Meetings.

Invited Conference Presentations

Indistinguishability of Linear Compartmental Models January 15, 2027
Identifiability and Indistinguishability in Dynamical Systems Chicago, Illinois
Joint Mathematics Meetings

Identifiability: Using Math and Trees to Solve Problems from Biology October 26-27, 2026
Mathematical Biology Special Session Davis, CA
Northern and Central California Section of SIAM Annual Meeting

Identifiability of Linear Compartmental Models July 20-23, 2026
Special Session on Biological Applications of Computer Algebra Waterloo, Ontario, Canada
International Congress on Mathematical Software

Identifiability of Linear Compartmental Models July 6-10, 2026
Special Session on Algebraic Biology Cleveland, OH
Society for Industrial and Applied Mathematics Annual Meeting

Identifiability: Using Math and Trees to Solve Problems from Biology August 6-9, 2025
MAA MathFest 2025 Sacramento, CA

Identifiability of Linear Compartmental Tree Models July 21-25, 2025
IMSI New Directions in Algebraic Statistics Chicago, IL

Indistinguishability of Linear Compartmental Models July 7-11, 2025
SIAM Conference on Applied Algebraic Geometry Madison, WI

Graph-Based Sufficient Conditions for Indistinguishability of Linear Compartmental Models May 3-4, 2025
AMS Western Sectional Meeting San Luis Obispo, CA

The Maximum Likelihood Degree of β-Stochastic Block Models January 8-11, 2025
AMS Joint Mathematics Meeting Seattle, WA

Identifiability of Linear Compartmental Models Using Graphs September 14-15, 2024
AMS Central Sectional Meeting University of Texas, San Antonio, TX

Identifiability of Linear Compartmental Models July 20-22, 2024
Workshop on Differential Algebra and Modeling North Carolina State University, Raleigh, NC

Graph-Theoretic Conditions for Identifiability of Linear Compartmental Tree Models May 4-5, 2024
AMS Western Sectional Meeting San Francisco State University, San Francisco, CA

Graph-Based Sufficient Conditions for Indistinguishability of Linear Compartmental Models (prepared but canceled) — January 3-6, 2024
AMS Joint Mathematics Meeting — San Francisco, CA

Graph-Based Sufficient Conditions for Indistinguishability of Linear Compartmental Models — October 7-8, 2023
AMS Central Sectional Meeting — Creighton University, Omaha, NE

Structural Identifiability of Series-Parallel LCR Systems — July 10-14, 2023
SIAM Conference on Applied Algebraic Geometry — Eindhoven University of Technology (virt.), Netherlands

Identifiable Linear Compartmental Tree Models — June 5-9, 2023
Differential Algebra and Related Topics XI — Queen Mary University London, United Kingdom

Identifiable Linear Compartmental Tree Models — May 16-20, 2022
Algebraic Statistics 2022 — University of Hawai'i at Manoa, Honolulu, HI

Colloquia and Non-Research Presentations

From Reflection to Readiness: ARC in Action — December 9, 2026
Mountain Pacific Association of Colleges and Employers — San Francisco, CA
with Alex Brown

Identifiability: Using Math and Trees to Solve Problems from Biology — November 13, 2026
Cal Poly San Luis Obispo Math Department Colloquium — San Luis Obispo, CA

From Reflection to Readiness: ARC in Action — October 24, 2026
California College Personnel Association — Fresno, CA
with Alex Brown

Identifiability: Using Math and Trees to Solve Problems from Biology — October 16, 2026
Fresno State Math Department Colloquium — Fresno State University, Fresno, CA

Cal-Bridge FAST Recruitment Presentations — September 24, October 6,8, 2026
MESA Workshops — Merced College (In Person) and Delta College (Virtual)
with Wing To

Your Leadership Identity Through Intentional Connections — March 28, 2026
Dr. Marvalene Hughes Leadership Conference — Stanislaus State, Turlock, CA
with Alex Brown

Indistinguishability of Linear Compartmental Models: The Mathematics of Uncertainty in Modeling — February 26, 2026
Stockton Talks Series — Stanislaus State, Turlock, CA

Standards-Based Grading in the Mathematics Classroom — January 31, 2026
2026 STEAM Symposium — Stanislaus State, Turlock, CA

Integrating Career and Professional Development in the Classroom — April 14, 2025
Faculty Center for Excellence in Teaching and Learning Workshop — Stanislaus State, Turlock, CA
with Alex Brown

The Complementary Functions of Instructors and Career and Professional Development Centers in Colleges March 14, 2025
Higher Education and Student Affairs in the Central Valley Conference with Alex Brown — Fresno, CA

Integrating Career and Professional Development in the Classroom March 1, 2025
Innovation in STEM Education Summit — Stanislaus State, Turlock, CA

The Complementary Functions of Instructors and Career and Professional Development Centers in Colleges October 26, 2024
California College Personnel Association (CCPA)'s 2024 Fall Institute — Merced, CA

Using Math and Trees to Solve Problems from Biology October 16, 2024
Algebra-Geometry-Combinatorics Seminar — San Francisco State University, San Francisco, CA

Using Math and Trees to Solve Problems from Biology September 4, 2024
M*A*T*H Colloquium — Sonoma State University, Rohnert Park, CA

The Complementary Functions of Instructors and Career and Professional Development Centers in Colleges August 7, 2024
All-Faculty Convocation State Center Community College District with Alex Brown — Fresno, CA

Using Math and Trees to Solve Problems from Biology May 28, 2024
Department of Operations Research Colloquium — Naval Postgraduate School, Monterey, CA

Using Math and Trees to Solve Problems from Biology April 11, 2024
Math Department Colloquium — CSU East Bay, Hayward, CA

Using Math and Trees to Solve Problems from Biology February 2, 2024
MATH Matters Speakers Series — Bakersfield College, Bakersfield, CA

Using Math and Trees to Solve Problems from Biology April 21, 2023
Math Department Colloquium — California State University, Long Beach (virt.)

Using Math and Trees to Solve Problems from Biology April 14, 2023
Math Department Colloquium — Morehead State University (virt.)

Using Math and Trees to Solve Problems from Biology October 11, 2022
Math and Computer Science Department Colloquium — Santa Clara University, Santa Clara, CA

Using Math and Trees to Solve Problems from Biology September 23, 2022
The Speaker Series for the Quantitatively Curious — Stanislaus State, Turlock, CA

Organizing and Facilitating

Cal-Bridge FAST Winter Workshop Lead December 2026
An introduction to research aimed at local community college students through a five-day intensive training, including background in Python, academic writing, and research topics in mathematics, computer science, physics, and astronomy.

Faculty Community of Practice (CoP) on Applied Student Research and Community Partnerships Fall 2026-Spring 2027
Biweekly meetings along with 11 faculty and various other campus partners exploring pinch points in developing and supporting community engaged learning and broader undergraduate research on campus.

CSU Quantitative Reasoning Center Committee on Academic Readiness Fall 2026-Present
Working with faculty from across the CSU to address undergraduate quantitative readiness.

The CSUBIOTECH Assistant Editor for Mathematics and Statistics Fall 2026-Present
The CSUBIOTECH is a new journal that provides a place for submission of articles related to the broad field of biotechnology.

Faculty Mentor Program Board of Directors Fall 2026-Present
The Faculty Mentor Program is designed to pair students, who are often first generation, with faculty mentors to support them throughout their college experience.

Cal-Bridge Steering Committee Fall 2026-Present
Cal-Bridge is a program that provides scholarship and professional development opportunities to undergraduate students interested in pursuing graduate school in Physics, Astronomy, Math, or Computer science.

Special Session on Algebraic Biology July 6-10, 2026
Organized a special session on Algebraic Biology at the 2026 Society for Industrial and Applied Mathematics Annual Meeting in Cleveland, Ohio.

Stan State Celebration of Teaching and Learning May 8, 2026
Organized and coordinated an end of year celebration bringing together faculty to discuss pedagogy, and announce the new director for the Faculty Center for Excellence in Teaching and Learning.

Instructional Institute Day January 23, 2026
Organized and coordinated a revitalized Instructional Institute Day, featuring faculty-led professional development workshops, an instructional resource fair, ADA Title II accessibility training, and community-building activities.

Credential Program Information Week December 1-5, 2025
Organized presentations from the coordinators of the Single Subject, Early Childhood Education credential, Internship programs.

Applying to PhD Programs October 22, 2025
Gave a presentation through the Career and Professional Development Center on applying to PhD programs aimed at undergraduate students.

California Forum for Diversity in Graduate Education October 18, 2025
Chaperoned several dozen underrepresented undergraduate students for Stanislaus State to the California Forum for Diversity in Graduate Education held at UC Merced.

Exploring Future Paths: Why would I want a PhD in Computational Biology? March 7, 2025
Two invited graduate students from the UC Berkeley computational biology program gave presentations and participated in a guided Q&A session for undergraduates at CSU Stanislaus interested in exploring graduate study in computational biology.

Stan State Summer Research Week 2023, 2024, 2025
Co-facilitated annual presentations including information about summer research programs and the application process, as well as guided Q&A sessions with several undergraduate students discussing their experiences summer research programs.

Math Superbowl and 6th Grade Math Blast Co-Organizer 2024, 2025, 2026
Co-organized the Math Superbowl (7th-12th grade) and 6th Grade Math Blast for local students to compete, including writing and editing various exams, trophy design and 3D-printing in 2025, and day-of preparation and execution.

New Faculty Orientation August 14-15, 2024
A two-day workshop introducing all new faculty to campus resources and requirements, along with information on inclusive pedagogical techniques.

How to Find Success In College English/Math/Science July 24-26, 2024
Three, 1.5 hour presentations on being successful in collegiate mathematics through the Health Careers Opportunity Pipeline Program with approximately 40 high school and community college students interested in pursing health-related careers.

Strategies for Student Engagement in the College of Science: An Open Exchange of Ideas May 9, 2024
Faculty development workshop funded by the Faculty Development Committee aimed at bringing together faculty to discuss inclusive strategies to increase student engagement in the classroom.

Career and Professional Development Faculty Learning Community Spring 2024 and Fall 2024
A Faculty Learning Community (FLC) related to career and professional development (supported by the University) to facilitate curricular innovation and a connection of faculty, students, and their coursework.

Special Session on Algebraic Approaches to Mathematical Biology January 3-6, 2024
An American Mathematical Society Special Session at the Joint Mathematics Meetings with invited speakers from across the country who specialize in applications of algebra and combinatorics to problems in biology.

The Speaker Series for the Quantitatively Curious Fall 2023 - Present
Co-facilitated biweekly presentations including faculty, members of the community, and students discussing the use of mathematics in their respective disciplines.

Workshops and Trainings

Mental Health First Aid Certified March, 2026
Nine-hour mental health first aid training — National Council for Mental Wellbeing

STEM Transfer Success Summit November 15, 2025
Stan State and local community college faculty and advisors — California State University, Stanislaus

Intro to Data Dashboards for Faculty October 8, 2025
Organized and attended. Presented by Dai Li of Stan State. — Virtual

Active Learning Community of Practice Fall, 2025
Biweekly meetings of Stan State faculty — Virtual

AI in the Classroom Community of Practice Fall, 2025
Biweekly meetings of Stan State faculty — Virtual

CSUBIOTECH Planning Meeting August 11, 2025
Faculty Consensus Group Representative — CSU Chancellor's Office, Long Beach, CA

MathFest August 6-9, 2025
Mathematical Association of America Sacramento, CA

Elevate Your Grading Workshop[s] March, April 2025
Division of Academic Affairs California State University, Stanislaus

CSUBIOTECH Planning Meeting August 5, 2024
Faculty Consensus Group Representative Sacramento State University

Gallo Future Leaders Symposium May 15, 2024
E. & J. Gallo Winery Modesto, CA

Transforming Student Experience & Success in Gateway STEM Courses Spring 2024
Faculty Learning Community with weekly meetings California State University, Stanislaus

CSU Symposium on Wellbeing-Informed Teaching March 15, 2024
California State University Chancellor's Office Virtual

Faculty Successful Grant Writing and Hands-on Workshop October 16-17, 2023
CSUBIOTECH California State University, Stanislaus

Transforming Hispanic Serving Institutions Workshop October 13, 2023
Warrior Cross Cultural Center California State University, Stanislaus

Research Experience for Undergraduate Faculty (REUF) Workshop August 7-11, 2023
Institute for Computational and Experimental Research in Mathematics (ICERM) Brown University

CSUBIOTECH Planning Meeting July 31, 2023
Faculty Consensus Group Representative CSU Chancellor's Office, Long Beach, CA

Social Justice in the Math Classroom - Putting Theory into Practice July 15, 17, 19, 21, 2023
Central California Math Project Summer Institute California State University, Stanislaus

CSU Student Success Dashboard Workshop April 26, 2023
Faculty Center for Excellence in Teaching and Learning California State University, Stanislaus (virt.)

Critical Issues in Mathematical Education: Mentoring for Equity Workshop March 22-24, 2023
Simons Laufer Mathematical Sciences Institute (formerly MSRI) Berkeley, CA

Becoming More Than the Sum of Our Parts Conference March 16-17, 2023
STEM-NET California State Polytechnic University, Pomona

Measures of Teaching Effectiveness Workshop October 27, 2022
Faculty Center for Excellence in Teaching and Learning California State University, Stanislaus

S-STEM Grant Planning Meeting October 11-12, 2022
Funded grant planning meeting California State University, Stanislaus

Using Inquiry to Teach Mathematics and Thinking September 29, 2022
Mathematical Association of America Workshop Virtual

MathFest/Project NExT Workshops August, 2022
- Active Learning with Active Calculus Mini-course (4 hours) Philadelphia, PA
- Teaching with Inquiry: Growing our Practice Together
- How to Create Safe and Community Responsive Outreach Programs
- Teaching and Leading Towards Institutional Change
- Making Teaching Matter More: Active-learning Strategies and Barriers to Implementation
- Townhall on Latinx Mathematicians
- The Role of Culturally Relevant Pedagogy and Identity Dev. in the Teaching and Learning of Math

Mentorship and Advising

Associated Students, Incorporated Fall 2025 - Present
Faculty Director of Associated Students, Incorporated.

McNair Scholars Program Research Mentor Spring 2025 - Spring 2026
Advising one undergraduate research student on a research project and graduate school applications over the course of the summer through the McNair Scholars Program.

Robert Noyce Teaching Scholarship Program Fall 2023 - Spring 2025
Mentoring two pre-service 6-12 teachers with biweekly meetings discussing organization and class preparation, pedagogical philosophy, and the credential program.

Cal-Bridge Fall 2023-Present
Co-mentoring two Cal-Bridge students (2023-2025 and 2025-present) with biweekly meetings discussing course selection, summer research experiences, and graduate school applications.

Math Club Faculty Advisor Fall 2023-Spring 2025, Fall 2026-Present
Co-advising the California State University, Stanislaus Math Club.

Math Major Advisor Fall 2023-Present
Advising California State University, Stanislaus math majors in course work and career preparation every semester (roughly 11-12 students per semester).

New Student Orientation 2023, 2025
Orientation advisor for four sessions of new student orientation, helping students choose classes and welcoming them to the university and the department.

Faculty Mentor Program Fall 2022-Present
Mentoring first-generation undergraduate students from a variety of majors. Through this program I have had four mentees so far.

Grants and Funding

CSUBIOTECH 2026 Curriculum Development Grant (Co-PI) Summer 2026

Title: *General Chemistry Workshop Curriculum to Improve Lower-Division STEM Student Success*
Overview: This proposal seeks to improve success and retention in General Chemistry I by developing and assessing a structured pre-course workshop for first- and second-year CSU students, particularly those from historically underserved backgrounds. Building on promising pilot results, the project will refine scalable curricular materials, evaluate student preparedness and academic outcomes, and support broader implementation of evidence-based chemistry support programs.
Budget: $14,968

CSUBIOTECH Travel Award (PI) Summer 2026

Overview: Travel funding to attend the Society for Industrial and Applied Mathematics (SIAM) Annual Meeting in Cleveland, Ohio to both organize a special session on Algebraic Biology and give a presentation in that session.
Budget: $1,500

McNair Undergraduate Student Research Project Summer 2025

Overview: Funding for myself and one undergraduate research student (Eduardo Chamorro) to do research over the summer on a project related to linear compartmental models.
Budget: ≈$5,800

Assigned Time for Exceptional Service to Students (PI) 2025-2026

Overview: Three units of release time awarded by the California State University Chancellor for mentoring several undergraduate research projects, helping organize junior high and high school math competitions, and redesigning the math degrees at CSU Stanislaus.
Budget: ≈$7,110

NSF Launching Early-Career Academic Pathways Grant (PI) 2025-2027

Title: *LEAPS-MPS: Exploring Properties of Linear Compartmental Models*
Overview: This project explores how mathematical diagrams called linear compartmental models (LCMs) can help researchers understand the structure and behavior of systems involving movement or flow, including how diseases spread, how nutrients cycle through ecosystems, or how medications are processed in the body. These models represent systems as interconnected compartments, with arrows indicating how quantities move between them. The investigator aims to identify when meaningful information about these systems, such as flow rates, can be reliably extracted from data, and when different models might appear identical in practice. The project's goal is to create simple, visual tests for analyzing these characteristics of LCMs, making the process more accessible and less computationally intensive. Housed at a Hispanic-Serving and Primarily Undergraduate Institution, this project provides undergraduates with opportunities to participate in cutting-edge mathematical research and equips them with background needed to pursue further education. A new colloquium series will further expand students' exposure to graduate-level research and support their preparation for advanced study.
Budget: $249,996

American Institute of Mathematics SQuaREs 2025-2027 (PI) Fall 2024

Title: *Maximum Likelihood Geometry for Statistical Network Models*
Overview: Our research team of six faculty from different, mostly primarily undergraduate institutions received full funding to spend one week per year at the American Institute of Mathematics (AIM) in Pasadena, California for three years. (Served as PI)
Budget: ≈$36,000

American Institute of Mathematics REUF Continuation Grant 2024 (PI) Summer 2024

Title: *Maximum Likelihood Degree of Stochastic Block Models*
Overview: Our research team of six faculty from different, mostly primarily undergraduate institutions received full funding to spend one week at the American Institute of Mathematics (AIM) in Pasadena, California to work on a project that started at the 2023 Research Experience for Undergraduate Faculty (REUF) at the Institute for Computational and Experimental Research in Mathematics (ICERM).
Budget: ≈$12,000

CSU Stanislaus Research, Scholarship, and Creative Activity Grant (PI) June 2024-June 2025

Title: *Standards-Based Grading Assessment Generating Application*
Overview: This internal research, scholarship, and creative activity grant is supporting PI Bortner's efforts to develop a web-based, artificial intelligence integrated application to generate exams using the Standards-Based Grading assessment technique. Support for this project comes in the form of 3 units of course release in the Fall 2024 semester, as well as the hiring of a software developer as a special consultant.
Budget: $10,000

CSU Stanislaus Faculty Development Committee Workshop Proposal (Co-PI) Spring 2024

Title: *Strategies for Student Engagement in the COS: An Open Exchange of Ideas*
Overview: In this round table discussion-style workshop, we discussed strategies for student engagement in STEM-related fields, including topics such as active learning, alternative grading and assessment strategies.
Budget: $1,250

CSU LSAMP Undergraduate Research Assistant Funding Fall 2023, Spring 2025 (x3)

Overview: Funded by the Louis Stokes Alliance for Minority Participation (LSAMP), four total students across were paid for engaging in undergraduate research.
Budget: $1,800 each (x4)

CSU Stanislaus College of Science Faculty Development Travel Grant

- Fall 2025 - Joint Mathematics Meetings, 2026 $2,000
- Fall 2024 - Joint Mathematics Meetings, 2025 $2,000
- Fall 2023 - Joint Mathematics Meetings, 2024 $2,000
- Spring 2023 - Simons Laufer Mathematical Sciences Institute Critical Issues in Math Education, $995
- Fall 2022 - Joint Mathematics Meetings 2023, $2,000

Fellowships

Career and Professional Development Center Faculty Fellow Fall 2023-Present
Faculty Fellowship with the Stanislaus State Career and Professional Development Center (CPDC) consulting with the Executive Director of the CPDC to develop an annual work plan that meets institutional needs, takes advantage of individual faculty strengths and interests, and sustains a manageable workload given the level of assigned time allotted to the position. Also, advising students for four hours per week on career readiness with an emphasis on graduate school.

CSUBIOTECH Faculty Consensus Group Representative Summer 2023-Present
One of four Stanislaus State CSUBIOTECH Faculty Consensus group representatives serving as a point-of-contact and mentor for campus grant applications, a reviewer for CSUBIOTECH grants and awards, committee member for program activities, and advisor on program priorities, policy, and governance.

California State University, Stanislaus Grant Writing Cohort Spring 2023-Fall 2023, Spring 2025
Weekly professional development meetings focused on all phases of grant application and implementation, with the ultimate goal of submitting an external grant application.

Mathematics Faculty Fellow: Catalyzing New Practices for the San Joaquin Valley to Innovate Effective Pedagogies in Lower-Division Mathematics and Chemistry Curriculum 2022-2023

Biweekly meetings focused on improving course outcomes in lower-division chemistry courses.

Mathematical Association of America Project NExT Fellow 2022-2023

Professional development program for new or recent Ph.D.s in the mathematical sciences addressing all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities. The program also provides the participants with a network of peers and mentors as they assume these responsibilities.

Reviewing and Judging

- 2026 Fall Cal-Bridge Research Symposium Feedback, September, 2026
- LEGO Build It Hackathon Judge, Computer Science Club, February, 2026
- NSF Grant Review Panel member for the STEM Education Directorate, 2026
- CSUBIOTECH 2026 Symposium Undergraduate Poster Judge, January, 2026
- CSUBIOTECH 2026 Symposium Undergraduate Poster Abstract Reviewer, Fall 2025
- Stanislaus State ASPIRE Hackathon Judge, April 25-27, 2025
- CSUBIOTECH 2025 Symposium Undergraduate Poster Abstract Reviewer, Fall 2024
- Stanislaus State ASPIRE Hackathon Judge, April 26-28, 2024
- Stanislaus State 38th Annual Undergraduate Research Competition Judge, March 8, 2024
- CSUBIOTECH 2024 Presidents' Commission Scholars Program Reviewer
- 2022 & 2023 CSU Mathematics Conference Session Reviewer

University/CSU Committees

- CSU Quantitative Reasoning Center Committee on Academic Readiness, 2026-Present [CSU]
- Stanislaus State Single Subject Credential Program Advisory Committee, 2026-Present
- Cal-Bridge Mathematics Steering Committee, 2026-Present [CSU]
- Academic Technology and Learning Committee, 2026-2028 [University]
- Committee on Committees, 2026-2027 [University]
- Faculty Center for Excellence in Teaching and Learning Director Search Committee, 2026 [University]
- Faculty Development Committee (co-chair/chair), 2024-2026 [University]

Department/College Committees

- College of Science Strategic Plan Committee, Fall 2026-Present [College]
- Math Masters Committee (chair), 2026-Present [Department]
- College of Science Curriculum Committee, 2024-Present [College]
- Committee on Committees, 2024-2025 [Department]
- Colloquium Committee, 2023-Present [Department]
- Math Major Revamp Committee (chair), 2023-Present [Department]
- Major Advising Committee, 2023-Present [Department]
- Graduate Approval Form Committee, 2023-Present [Department]
- Math Competition Planning Committee, 2023-Present [Department]
- Math/SIAM/PME Club Advisor, 2023-2025, 2026-Present [Department]

Academic Honors and Awards

- **Winner** of the Elizabeth Anne B. Papageorge Faculty Development Award, 2025-2026
- Nominated for the Outstanding Service in Faculty Governance Award, 2025-2026
- Nominated for the Elizabeth Anne B. Papageorge Faculty Development Award, 2023-2024
- Winner of the North Carolina State University Franke Norris Griggs Award for Service, 2022
- University of Nebraska-Lincoln Chair's Prize for Outstanding Graduating Undergraduate, 2017
- University of Nebraska-Lincoln Parents' Recognition Award for Outstanding Faculty and Staff, 2017

Teaching Experience (California State University, Stanislaus)

MATH 4980: Independent Study on 3D Modeling and Design	Fall 2026 (1 student)
MATH 3400: Introduction to Proofs	Fall 2026 (26 students)
MATH 1073: Supporting College Algebra	Fall 2026 (51 students, 2 sections)
MATH 4980: Independent Study on Algebraic Geometry	Spring 2026 (2 students)
MATH 3060: Modern Geometry	Spring 2026 (23 students)
MATH 1422: Calculus II Lab	Spring 2026 (12 students)
MATH 1412: Calculus I Lab	Spring 2026 (12 students)
MATH 4540: Abstract Algebra II	Fall 2025 (6 students)
MATH 2300: Discrete Structure	Fall 2025 (31 students)
MATH 4530: Abstract Algebra	Spring 2025 (26 students)
MATH 1422: Calculus II Lab	Spring 2025 (7 students)
MATH 1420: Calculus II	Spring 2025 (28 students)
MATH 3400: Set Theory and Logic	Fall 2024 (21 students)
MATH 1422: Calculus II Lab	Fall 2024 (10 students)
MATH 1070: College Algebra	Fall 2024 (33 students)
MATH 4980: Independent Study on Machine Learning with Graphs	Spring 2024 (7 students)
MATH 3600: Theory of Numbers	Spring 2024 (25 students)
MATH 2460: Introduction to Differential Equations	Spring 2024 (27 students)
MATH 3400: Set Theory and Logic	Fall 2023 (19 students)
MATH 3400: Set Theory and Logic	Spring 2023 (21 students)
MATH 2300: Discrete Structures	Spring 2023 (27 students)
MATH 1070: College Algebra	Spring 2023 (33 students)
MATH 1420: Calculus II	Fall 2022 (16 students)
MATH 1410: Calculus I	Fall 2022 (24 students)

Professional Memberships

- American Mathematical Society (AMS), 2022-Present
- Mathematical Association of America (MAA), 2019-Present
- Society for Industrial and Applied Mathematics (SIAM), 2019-Present